The `dihiggs` model

A smooth, exactly Bayesian model of the HEFT di-Higgs mass spectrum, and its systematic uncertainty

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Abstract

This note describes, step by step, the model of the gluon-fusion di-Higgs cross section implemented in `dihiggs/smooth.py`. It replaces the October 2026 version of this note, which described the neural network `HEFTNet`; the network is now a cross-check (section 9). The model predicts the density $d\sigma/dm_{hh}$ as a function of the di-Higgs mass m_{hh} and of five HEFT Wilson coefficients. The dependence on the coefficients is a known polynomial with 23 terms. Each of the 23 unknown functions of m_{hh} that multiply those terms is a smooth spline. Because the model is linear in its parameters, its posterior density is *exactly* Gaussian. No training, ensembles or sampling are needed. The note gives particular attention to the model's *systematic uncertainty*: an uncertainty that comes not from statistical fluctuations but from an imperfection of the physics description. Here the imperfection is that the POWHEG spectra depart slightly from the 23-term form that the theory says they must have (section 6).

In brief

1. **The model.** $\dfrac{d\sigma}{dm_{hh}}(m;\theta) = \sum_{i=1}^{23} C_i(\theta)\, a_i(m)$, where the $C_i(\theta)$ are the 23 known HEFT monomials and $a_i(m) = \phi(m)\sum_{j=1}^{40}\beta_{ij}B_j(m)$, with B_j cubic B-splines and ϕ the two-body threshold factor. The 920 coefficients β_{ij} are fitted to the *bin integrals* of 3,989 POWHEG spectra by Poisson maximum likelihood, with a smoothness prior whose strengths are set by the evidence.
2. **Three uncertainties** enter any comparison with POWHEG: the Poisson noise of the POWHEG sample (statistical, $\approx$ 1,000 events per bin near threshold); the fit uncertainty of the model (statistical, a median 2.9% of the noise); and the **model-error term** (systematic).
3. **The systematic.** POWHEG departs from the 23-term form smoothly in m_{hh}, and most where the 23 terms cancel. We model the departure as a relative error $\varepsilon(m)$ on each term, giving a standard deviation $\delta = \varepsilon(m)\sum_i |C_i(\theta)A_i|$ per bin. ε is measured on data not used in the fit: 0.08% at threshold, 0.28% at 600 GeV, 3.7% at 1.45 TeV (three parameters in all).
4. **Validation** on 1,000 independent spectra. The accuracy equals that of the best possible model of the 23-term form (likelihood-ratio statistic $-2\ln\lambda$ per bin 1.566 below and 1.100 above 555 GeV). With the systematic included, the pulls have standard deviation 1.004 and are calibrated at every degree of cancellation.

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1 What we want to model, and why

In gluon fusion, $gg \to hh$, new physics described by HEFT changes both the size and the shape of the di-Higgs invariant-mass spectrum. The relevant part of the HEFT Lagrangian is [1]

$$\mathcal{L}_{\text{HEFT}} \supset -m_t\Big(c_t\,\frac{h}{v} + c_{tt}\,\frac{h^2}{v^2}\Big)\bar{t}t - \kappa_\lambda\,\frac{m_h^2}{2v}\,h^3 + \frac{\alpha_s}{8\pi}\Big(c_{ggh}\,\frac{h}{v} + c_{gghh}\,\frac{h^2}{v^2}\Big)G^a_{\mu\nu}G^{a,\mu\nu}, \tag{1}$$

where $v \approx 246$ GeV. Each coefficient has a simple meaning:

coefficient	interaction it controls	SM value
$\kappa_\lambda \equiv c_{hhh}$	Higgs self-coupling hhh	1
c_t	top Yukawa coupling $t\bar{t}h$	1
c_{tt}	contact coupling $t\bar{t}hh$	0
c_{ggh}	effective (point-like) coupling ggh	0
c_{gghh}	effective (point-like) coupling $gghh$	0

We collect them into $\theta = (\kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh})$.

We want the density $d\sigma/dm_{hh}(m;\theta)$, valid everywhere in a broad region of the five-dimensional θ space, together with an honest uncertainty. Such a function lets us *reweight* existing simulated events to any other θ, instead of running a new simulation for every point. It is also an ingredient of the simulation-based construction of confidence sets in θ that motivates this project [2, 3].

Figure 1 shows four of the simulated spectra. They are very different: their sizes span three orders of magnitude, some fall steeply from threshold, one peaks near 700 GeV, and spectrum (b) has a sharp dip near 330 GeV. The dip is a sign of destructive interference between different contributions to the amplitude. The model has to describe all of these with one set of parameters. Interference will matter again when we come to the systematic uncertainty (section 6).

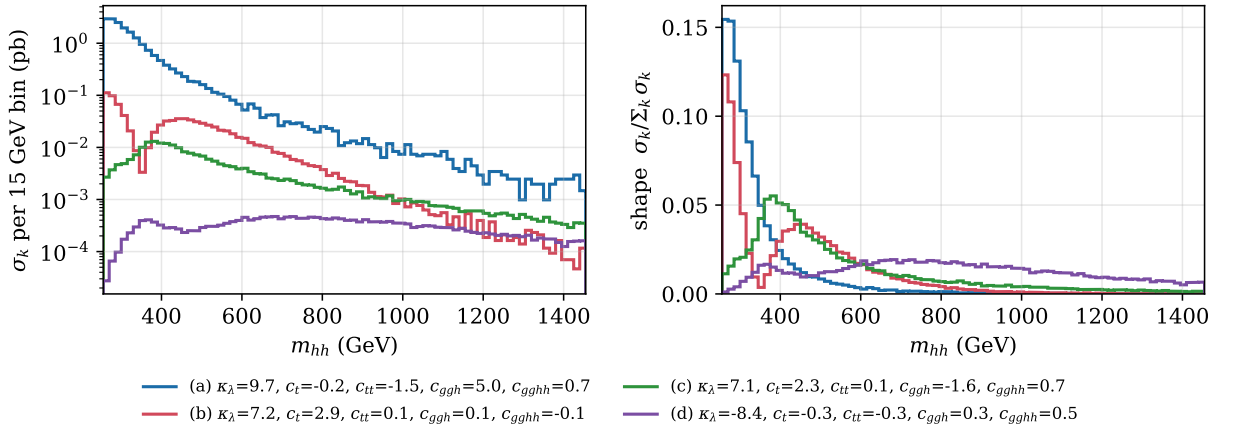


Figure 1: Four POWHEG spectra from the held-out test set, chosen to show the range of shapes. Left: cross section per 15 GeV bin (log scale). Right: the same spectra divided by their totals, so that only the shapes are compared.

2 The physics: why σ is a polynomial in θ

2.1 The amplitude is linear in the couplings

At leading order (LO) there are five classes of Feynman diagrams for $gg \to hh$ in HEFT (fig. 2). Each class contains exactly one combination of couplings: two top Yukawa vertices in the box, a Yukawa vertex and the self-coupling in the top-loop triangle, and so on. Therefore the full amplitude is a *linear* combination of five fixed sub-amplitudes,

$$\mathcal{M} = c_t^2\,\mathcal{M}_1 + c_t\kappa_\lambda\,\mathcal{M}_2 + c_{tt}\,\mathcal{M}_3 + c_{ggh}\kappa_\lambda\,\mathcal{M}_4 + c_{gghh}\,\mathcal{M}_5 \;\equiv\; \sum_{i=1}^{5} v_i(\theta)\,\mathcal{M}_i, \tag{2}$$

with

$$\boldsymbol{v}(\theta) = \big(c_t^2,\ c_t\kappa_\lambda,\ c_{tt},\ c_{ggh}\kappa_\lambda,\ c_{gghh}\big). \tag{3}$$

The sub-amplitudes $\mathcal{M}_i$ depend on the kinematics (on m_{hh} and the scattering angle) but *not* on θ. All of the θ dependence is in the five numbers v_i.

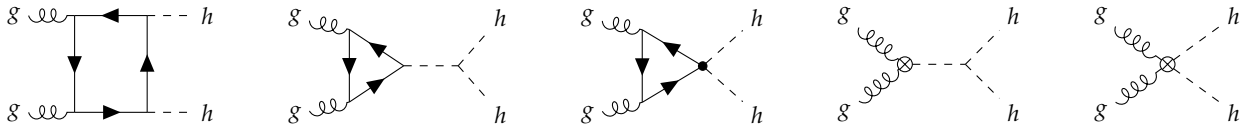


Figure 2: The five classes of LO diagrams for $gg \to hh$ in HEFT and the coupling combination each one carries. Solid lines are top quarks, dashed lines Higgs bosons; the dot is the $t\bar{t}hh$ contact vertex and the crossed dots are the point-like gluon–Higgs vertices of eq. (1).

2.2 Squaring the amplitude gives 15 terms

A cross section is proportional to $|\mathcal{M}|^2$. Squaring eq. (2) gives

$$|\mathcal{M}|^2 = \sum_{i=1}^{5}\sum_{j=1}^{5} v_i\, v_j\, \mathrm{Re}\big(\mathcal{M}_i \mathcal{M}_j^*\big). \tag{4}$$

Now integrate over everything except m_{hh} (the scattering angle, and the parton distributions). Because the v_i do not depend on the kinematics, they come out of the integral:

$$\frac{d\sigma}{dm_{hh}}(m_{hh},\theta) = \sum_{i\le j} v_i(\theta)\, v_j(\theta)\, K_{ij}(m_{hh}), \qquad K_{ij}(m_{hh}) \propto \int \mathrm{Re}\big(\mathcal{M}_i \mathcal{M}_j^*\big)\, d(\text{everything but } m_{hh}). \tag{5}$$

K is a symmetric 5×5 matrix of functions of m_{hh}, so it has $5\cdot 6/2 = 15$ independent entries. The 15 products $v_i v_j$, written out in the couplings, are the first 15 monomials in table 4. The diagonal entries K_{ii} are squared magnitudes and so are non-negative; the off-diagonal entries are *interference* terms and can have either sign. Negative interference is what produces dips such as the one in spectrum (b) of fig. 1.

2.3 Next-to-leading order adds 8 more terms

The training data were generated with POWHEG at next-to-leading order (NLO) in QCD. At NLO new diagram classes appear, and the polynomial grows from 15 to 23 terms [1]. One way to see where the 8 extra terms come from is to consider the NLO diagrams in which each Higgs boson attaches to the gluons through its own single-Higgs vertex, either a top loop ($\propto c_t$) or the point-like ggh vertex ($\propto c_{ggh}$). Their coupling factors are $\boldsymbol{w} = (c_t^2,\ c_t c_{ggh},\ c_{ggh}^2)$. Interfering these with the LO amplitude gives products $w_a v_i$. The products with $w_1 = c_t^2$ reproduce LO monomials, but those with $c_t c_{ggh}$ and c_{ggh}^2 give exactly the 8 new monomials 16–23 of table 4.

Key idea

For *every* m_{hh} the density is a fixed polynomial in θ:

$$\boxed{\frac{d\sigma}{dm_{hh}}(m;\theta) = \sum_{i=1}^{23} C_i(\theta)\, a_i(m) = \boldsymbol{C}(\theta)\cdot\boldsymbol{a}(m)} \tag{6}$$

The 23 monomials $C_i(\theta)$ are *known exactly* (table 4). Only the 23 functions $a_i(m)$, in pb/GeV, are unknown. They are the integrated, NLO-corrected interference kernels K_{ij}. The coupling dependence and the kinematic dependence *factorise*.

When is the factorised form exact?

The derivation needs only two things. First, the amplitude must be linear in the five coupling combinations of eq. (3), together with their NLO extensions. Second, nothing else in the calculation may depend on θ: the renormalisation and factorisation scales, the parton distributions and the phase-space cuts must be the same for every θ. In a fixed-order calculation that meets these conditions, eq. (6) is exact, not an approximation. This matters later: if the POWHEG spectra depart from the 23-term form, the departure tells us about the calculation or its numerical implementation, not about missing physics in the form itself (section 6).

The LHC Higgs Working Group determines the a_i (called dA_i there) one bin at a time with a least-squares fit [1]. Here they are smooth functions of m_{hh}, determined by all bins at once.

3 The data: POWHEG spectra are histograms of counts

Spectra. The file `data/powheg_total_gauss_params_13TeV.csv.gz` contains 4,999 POWHEG NLO spectra at $\sqrt{s} = 13$ TeV. Each row holds one parameter point θ and the cross section in pb in 100 bins of 15 GeV ($[15k,\ 15(k+1))$ GeV). The points are scattered roughly as Gaussians centred on the SM point, with standard deviations of about 5 for κ_λ and about 2 for the other four coefficients. We use the 80 bins from 255 to 1455 GeV. A thousand spectra are set aside untouched as the *test* spectra. The remaining spectra are "diced" into one row per bin (θ_s, k, t_{sk}) and shuffled. 88% of these rows (281,044, from 3,989 spectra, including 1,814 restored empty bins) are used in the fit. The other 12%, the 38,076 *quick-test* rows, are never used in the fit and serve to measure the systematic uncertainty (section 6.5).

Exact counts. Every spectrum is a histogram of 40,000 generated events of equal weight q_s (a few incomplete runs have fewer; `data/README.md`). The cross section in bin k of spectrum s is therefore *exactly*

$$t_{sk} = q_s\, n_{sk}, \qquad n_{sk} \in \{0, 1, 2, \dots\}, \tag{7}$$

and `dihiggs/counts.py` recovers q_s and the integer counts n_{sk} from the stored numbers by a greatest-common-divisor search (`docs/counts/counts_note.pdf`). The counts are Poisson distributed and, apart from a negligible multinomial correlation, independent. Near threshold a bin holds about 1,000–1,800 events; at 1.4 TeV, about 20. This gives us the exact statistical noise of every number we fit, which is what makes the separation of statistical and systematic uncertainty in section 6 possible.

4 The model

4.1 Smooth coefficient functions

Each coefficient function is a sum of $J = 40$ cubic B-splines times a threshold factor:

$$a_i(m) = \phi(m) \sum_{j=1}^{J} \beta_{ij}\, B_j(m), \qquad \phi(m) = \sqrt{1 - \frac{4m_h^2}{m^2}}, \quad m_h = 125\ \text{GeV}. \tag{8}$$

B-splines in brief

A cubic B-spline $B_j(m)$ is a smooth bump. It is a piecewise cubic polynomial that is non-zero only over four consecutive knot intervals, and its pieces join with continuous first and second derivatives. A weighted sum of B-splines can represent any smooth curve whose wiggles are not finer than the knot spacing. Each coefficient β_{ij} acts locally: it moves a_i only near knot j. B-splines are the standard tool for smoothing in statistics [4, 5].

Knots. The functions change fastest just above threshold and slowly above 600 GeV. The 38 breakpoints are therefore spaced uniformly in $\sqrt{m - 250\ \text{GeV}}$ between 250 and 1500 GeV, which makes them dense just above threshold (fig. 3). The knot vector is *clamped*: each end point is repeated three more times. As a result the 40 basis functions sum to one everywhere on $[250, 1500]$ GeV, and only B_1 is non-zero at 250 GeV.

Threshold factor. At $m_{hh} = 2m_h$ the two-body phase space vanishes like the Higgs velocity in the hh rest frame, $\phi = \sqrt{1 - 4m_h^2/m^2}$, and so does $d\sigma/dm_{hh}$. Building ϕ in leaves the splines only the smooth remainder to describe. Within 255–1455 GeV the factor hardly changes the fit. Below 255 GeV it is decisive: predicting the unused 240–255 GeV bin, the model with ϕ is 8% high, the model without it 55% high.

4.2 The prediction is a bin integral, and it is linear

POWHEG gives the integral of the density over each bin. The model's prediction for bin $k = [e_k, e_{k+1}]$ of a spectrum at θ is therefore the integral of eq. (6) over the bin, not its value at the centre:

$$f_k(\theta) = \int_{e_k}^{e_{k+1}} \frac{d\sigma}{dm}\, dm = \sum_{i=1}^{23} C_i(\theta)\, A_{ik}, \qquad A_{ik} = \int_{e_k}^{e_{k+1}} a_i\, dm = \sum_{j=1}^{J} \beta_{ij}\, I_{jk}, \qquad I_{jk} = \int_{e_k}^{e_{k+1}} \phi\, B_j\, dm. \tag{9}$$

The numbers I_{jk} depend only on the knots and the binning. They are computed once by 16-point Gauss–Legendre quadrature in each bin (`SplineBasis.integrals`). The prediction is *linear* in the $23J = 920$ coefficients β_{ij}. Everything that follows (an exact posterior, exact predictions for any binning, an exact evidence) rests on this linearity.

The free fit as a limit. If each basis function is a box covering one bin, and $\phi = 1$, then eq. (9) becomes a separate fit of 23 numbers per bin, $23 \times 80 = 1{,}840$ in all. This is the Working Group's per-bin approach [1]. No model of the form eq. (6) can describe the data better than this free fit, so we use it as a benchmark. It reappears in section 6 as the key piece of evidence.

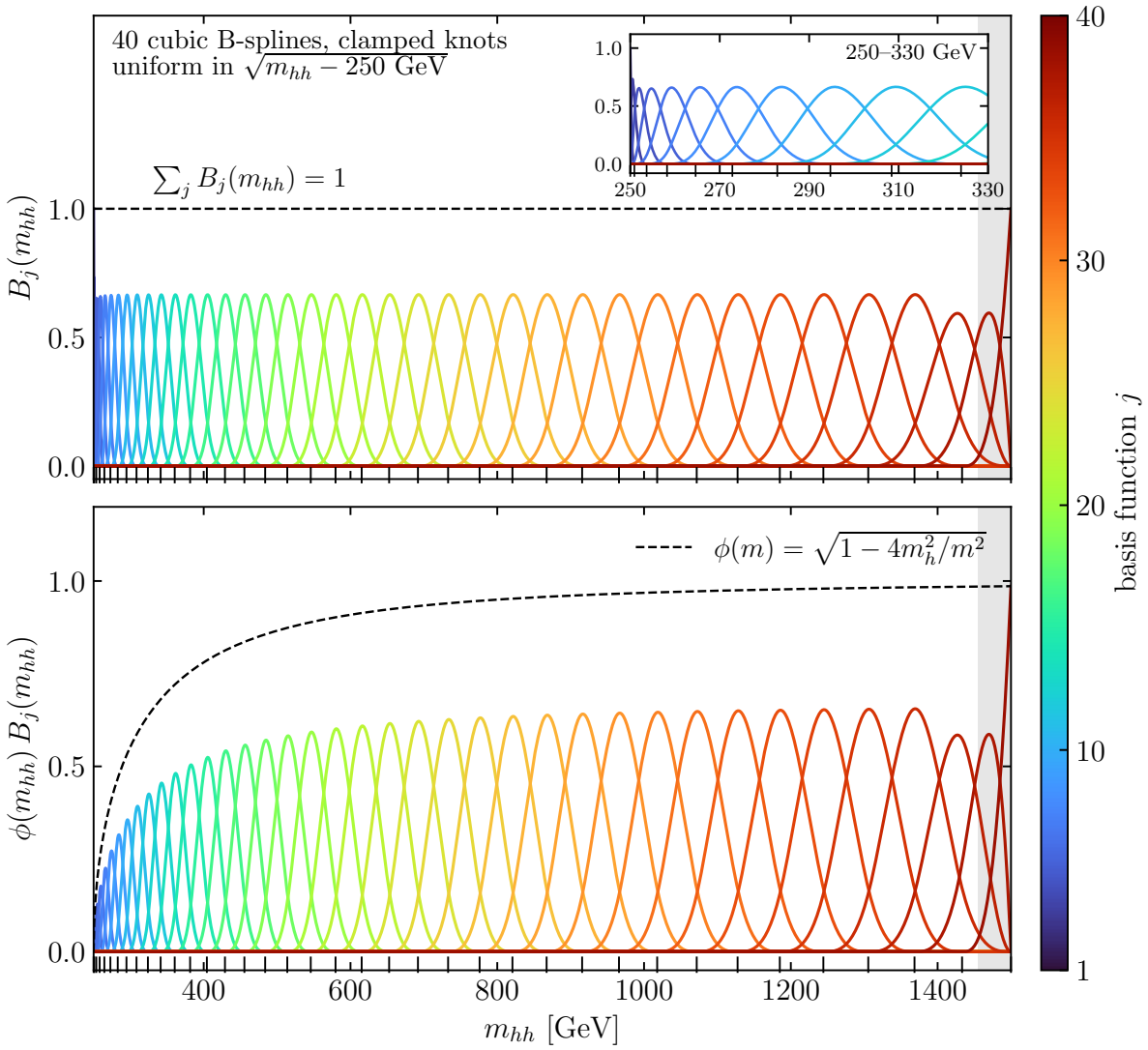


Figure 3: The basis of the smooth model. Top: the 40 cubic B-splines $B_j(m_{hh})$, coloured by index j. The breakpoints (ticks) are uniform in $\sqrt{m_{hh} - 250\ \text{GeV}}$, so they are dense just above threshold (inset), where the spectra change fastest. The knot vector is clamped (each end point repeated three more times), so the B_j sum to one everywhere on $[250, 1500]$ GeV and only B_1 (B_{40}) is non-zero at the lower (upper) end. Bottom: the basis functions $\phi(m_{hh})B_j(m_{hh})$ multiplied by the coefficients β_{ij} in $a_i(m_{hh}) = \phi(m_{hh})\sum_j \beta_{ij} B_j(m_{hh})$, with the threshold factor $\phi = \sqrt{1 - 4m_h^2/m_{hh}^2}$ (dashed). The model is used only below 1455 GeV; the shaded region is beyond the last bin.

4.3 Parameter count

quantity	number	determined by
spline coefficients β_{ij}	920	the fit (section 5)
smoothness strengths λ_i	23	the evidence
model-error coefficient $\varepsilon(m)$	3	the quick-test rows (section 6.5)
total	946	

The data determine about 602 of the 920 linear combinations of the β_{ij} (the "number of well-determined parameters" γ [6]). The rest are fixed by the smoothness prior. For comparison, `HEFTNet` has 3,944 parameters and the free fit 1,840.

5 The fit

5.1 Likelihood: weight by the model, not by the data

With hundreds of events in most bins a Gaussian likelihood is accurate:

$$-\ln p(\boldsymbol{t} \mid \boldsymbol{\beta}) = \tfrac{1}{2} \sum_{s,k} \frac{(t_{sk} - f_{sk})^2}{s^2_{\text{noise}}(s,k)} + \text{const}, \qquad s^2_{\text{noise}}(s,k) = q_s \, \max(f_{sk}, \, q_s). \tag{10}$$

The variance is the Poisson variance *of the model's prediction*, $q_s^2 \, \mu_{sk}$ with $\mu_{sk} = f_{sk}/q_s$ the predicted number of events. The max protects the few bins where the prediction is below one event. Weighting by the *observed* count instead (Neyman's χ^2, $s^2_{\text{noise}} = q_s t_{sk}$) gives downward fluctuations larger weights. It biases the fit low by about one event per bin [7], which, with 3,989 spectra pooled and 20–50 events per bin at high mass, amounts to many standard deviations. Because s^2_{noise} depends on the fit, the fit is iterated: fit, recompute the variances, refit. Each step moves half-way, by the geometric mean of the old and new variances, to avoid oscillation (`SmoothHEFT.fit_poisson`, 12 iterations). At convergence this is the Poisson maximum-likelihood fit. A toy with known coefficient functions confirms that the fit recovers them within its quoted uncertainties (`code/smooth_toy.py`: pulls with mean $+0.06$ and s.d. 0.97).

5.2 Smoothness prior and the evidence

For each function i, a Gaussian prior on the second differences of its coefficients (a "P-spline" [5]):

$$-\ln \pi(\boldsymbol{\beta}_i \mid \lambda_i) = \tfrac{1}{2} \lambda_i \sum_{j=3}^{J} \left(\beta_{ij} - 2\beta_{i,j-1} + \beta_{i,j-2}\right)^2 + (\text{negligible ridge}) + \text{const}. \tag{11}$$

This is the discrete analogue of a penalty on the curvature $\int (a_i'')^2 \, dm$. Straight lines are not penalised. The 23 strengths λ_i, one per function because the functions differ by orders of magnitude in size, are chosen by maximising the evidence $p(\boldsymbol{t} \mid \boldsymbol{\lambda})$ with MacKay's fixed-point updates [6]. Because the model is linear, the evidence is available in closed form.

5.3 The exact posterior

Likelihood and prior are both Gaussian in $\boldsymbol{\beta}$, so the posterior is exactly Gaussian:

$$\boldsymbol{\beta} \mid \boldsymbol{t} \;\sim\; \mathcal{N}\big(\hat{\boldsymbol{\beta}}, \; H^{-1}\big), \qquad H = \Phi^{\mathsf{T}} W \Phi + \Lambda, \qquad \hat{\boldsymbol{\beta}} = H^{-1} \Phi^{\mathsf{T}} W \boldsymbol{t}, \tag{12}$$

where each row of Φ is $\boldsymbol{C}(\theta_s) \otimes \boldsymbol{I}_k$ (eq. (9)), $W = \text{diag}(1/s^2_{\text{noise}})$, and Λ is the prior precision. The factorisation into monomials and bins makes $\Phi^{\mathsf{T}} W \Phi = \sum_{k=1}^{80} S_k \otimes \boldsymbol{I}_k \boldsymbol{I}_k^{\mathsf{T}}$, with $S_k = \sum_s w_{sk} \boldsymbol{C}_s \boldsymbol{C}_s^{\mathsf{T}}$ a 23×23 matrix, so the whole fit takes about a minute on a laptop CPU.

Predictions. For any θ and any interval, the prediction is $\boldsymbol{g}^{\mathsf{T}} \hat{\boldsymbol{\beta}}$ and its *fit variance* is $s^2_{\text{fit}} = \boldsymbol{g}^{\mathsf{T}} H^{-1} \boldsymbol{g}$, with $\boldsymbol{g} = \boldsymbol{C}(\theta) \otimes \int \boldsymbol{\phi} \boldsymbol{B} \, dm$ over that interval. Correlations between bins, between values of θ, and between the a_i come out automatically. The fit uncertainty is small, a median 2.9% of the Poisson noise of a POWHEG bin: the 3,989 training spectra determine the 23 functions very well. This is *not* the dominant uncertainty, as the next section shows.

6 The systematic uncertainty: the model-error term

6.1 Three kinds of uncertainty

Compare the model's prediction f_{sk} with the POWHEG cross section t_{sk} in one bin of one spectrum. Three distinct effects make them differ:

uncertainty	kind	origin	size (median, relative to the noise)
Poisson noise, s_{noise}	statistical	finite number of events in the POWHEG sample being compared; belongs to the *reference*, not to the model	1 (by definition)
fit uncertainty, s_{fit}	statistical	finite number of training spectra; the posterior width H^{-1}	0.03
model-error term, δ	systematic	POWHEG does not exactly obey the 23-term form of eq. (6)	0.17 below and 0.16 above 555 GeV; up to 1 where the 23 terms cancel

The first two are *statistical*: they come from random fluctuations, they shrink as samples grow, and we know their size exactly (sections 3 and 5). The third is a *systematic* uncertainty in the sense physicists use the word. It is not caused by statistical fluctuations. It is caused by a deficiency in the physics models, here in the agreement between the POWHEG calculation and the functional form we impose on it. More training spectra would not reduce it. The three add in quadrature in the pull,

$$z_{sk} = \frac{t_{sk} - f_{sk}}{\sqrt{s^2_{\text{noise}} + s^2_{\text{fit}} + \delta^2}}, \tag{13}$$

which should be a standard normal variable if all three are right.

What this systematic is *not*

The model is built to reproduce POWHEG. Its systematic uncertainty measures how well *POWHEG agrees with the 23-term form*, i.e. the internal consistency of the simulated spectra. It does *not* include the uncertainties of the NLO prediction relative to nature: renormalisation and factorisation scale variations, parton distributions, the top-mass scheme, missing higher orders. In a physics analysis those must be added separately, as for any generator.

6.2 The evidence: an accuracy floor that no model of the 23-term form can cross

The floor. A natural measure of how well a prediction μ (in events) describes an observed count n is the likelihood-ratio statistic

$$-2\ln\lambda = -2\ln\frac{\text{Poisson}(n;\mu)}{\text{Poisson}(n;n)} = 2\Big[\mu - n + n\ln\frac{n}{\mu}\Big]. \tag{14}$$

For a correct prediction it follows, asymptotically, a χ^2 distribution with one degree of freedom (Wilks). No parameter is fitted to the test spectra, so its mean per bin should be 1. On the 1,000 test spectra it is 1.566 between 255 and 555 GeV and 1.100 above, for the smooth model. For the *free fit*

(23 independent numbers per bin, the best that any model of the form eq. (6) can do) it is 1.559 and 1.105. For the network ensemble it is 1.561 and 1.121 (table 3). Three very different models reach the same floor. The excess over 1 is therefore not a property of any particular model of the a_i. It is a property of the data: the POWHEG spectra do not exactly have the 23-term form.

What the floor is not. The excess is not small-count statistics (near threshold the bins hold over 1,000 events, and Neyman, Pearson and likelihood-ratio measures agree). It is not the normalisation: the totals scatter about the fit by 0.47%, as expected from the counts alone (0.51%). And it is not the fit uncertainty, which is 3% of the noise (`docs/modelerror/model_error_note.pdf`).

It grows with cancellation. Define for each spectrum the *cancellation ratio*

$$R_s = \frac{\sum_k t_{sk}}{\sum_k \sum_i |C_i(\theta_s) A_{ik}|}, \tag{15}$$

the total cross section divided by the sum of the magnitudes of its 23 terms. $R = 1$ when all terms have the same sign. $R \ll 1$ when the cross section is a small difference of large terms, i.e. when the interference is strongly destructive. Divide the 1,000 test spectra into fifths of R. Table 1 shows that the excess, measured by $\langle z^2 \rangle$ with only the statistical uncertainties, sits almost entirely in the most cancelling fifth.

cancellation fifth range of R	1 (strongest) 0.002–0.12	2 0.12–0.23	3 0.23–0.42	4 0.42–0.73	5 (weakest) 0.73–1.00
$\langle z^2 \rangle$, noise and fit only	1.70	1.10	1.07	1.06	1.01
neighbour correlation, 255–555 GeV	0.60	0.14	0.08	0.04	0.01
neighbour correlation, $>$ 555 GeV	0.23	0.04	0.04	0.01	-0.01

Table 1: The excess and the smoothness of the residuals, by degree of cancellation (200 test spectra per column). Residuals are $(t - f)/s_{\text{noise}}$; "neighbour correlation" is the correlation between the residuals of adjacent bins of the same spectrum. Poisson noise alone gives $\langle z^2 \rangle = 1$ and no correlation between bins.

It is smooth in m_{hh}. Poisson noise is independent from bin to bin. The residuals are not: in the most cancelling fifth, neighbouring bins below 555 GeV are correlated by 0.60 (table 1), and the correlation extends over 5–8 bins. POWHEG departs from the 23-term form by a *smooth*, spectrum-dependent amount. Figure 4 shows three typical cases. With little cancellation POWHEG and the model agree to within Poisson noise. With strong cancellation the residuals form a smooth wave that the 23-term form cannot follow. The most cancelling test spectrum (ID 917, $R = 0.0025$: its cross section is a quarter of a per cent of the sum of the magnitudes of its terms) has a mean $-2 \ln \lambda$ of 51 per bin.

6.3 Why errors grow when terms cancel

Suppose that in the POWHEG calculation for spectrum s each of the 23 terms carries a small relative error $\delta_i(m; s)$:

$$\frac{d\sigma_s^{\text{POWHEG}}}{dm} = \sum_{i=1}^{23} C_i(\theta_s)\, a_i(m)\, \left[1 + \delta_i(m; s)\right] \quad \Longrightarrow \quad \text{error} = \sum_{i=1}^{23} C_i(\theta_s)\, a_i(m)\, \delta_i(m; s). \tag{16}$$

Two features of eq. (16) explain everything in section 6.2.

1. **Amplification by cancellation.** The error is of the order of $\varepsilon \sum_i |C_i a_i|$, where ε is the typical size of the δ_i, while the cross section is $\sum_i C_i a_i$. The *relative* error on the cross section is therefore about ε / R. An error of 0.5% on terms that cancel to 0.25% of their size is a 200% error on the result. When the terms add ($R \approx 1$) the same error is 0.5%, far below the Poisson noise.

2. **The errors must differ from spectrum to spectrum.** If the $\delta_i(m)$ were the same in every spectrum, the error would be $\sum_i C_i(\theta)\,[a_i \delta_i]$, which is again of the 23-term form with $a_i \to a_i(1+\delta_i)$. The fit would simply absorb it, and no floor would appear. A floor can arise only if the errors vary from run to run, i.e. with s, while being smooth in m within each run. This is what one expects if each HEFT point was generated as an independent run, with its own integration grids and numerical errors.

An analogy from the bench

Measure a small quantity as the difference of two large readings from instruments calibrated to 0.5%. A 0.5% calibration error on each reading is harmless for a large sum but ruinous for a small difference. Moreover, if the same instruments are used every time, the calibration error is a fixed offset that can be measured and removed. Only if the calibration changes from one measurement to the next does it become an uncertainty. That is the situation here: the "instruments" are the 23 components of the POWHEG calculation, and the "calibration" is apparently redone with every run.

6.4 The model of the systematic

Equation (16) suggests modelling the systematic as a random error, smooth in m, of relative size $\varepsilon(m)$ on each of the 23 terms. How the δ_i of one spectrum are correlated among themselves is unknown. Two limits bracket the possibilities. For the standard deviation of $\sum_i C_i A_{ik} \delta_i$ in bin k, with A_{ik} the bin integral of a_i:

$$\text{independent } \delta_i: \quad \delta_{sk} = \varepsilon(m_k) \Big[\sum_{i=1}^{23} \big(C_i(\theta_s) A_{ik} \big)^2 \Big]^{1/2}, \tag{17}$$

$$\text{any correlation (upper bound):} \quad \delta_{sk} = \varepsilon(m_k) \sum_{i=1}^{23} \big| C_i(\theta_s) A_{ik} \big|. \tag{18}$$

The second line follows from the triangle inequality, since the standard deviation of a sum is at most the sum of the standard deviations. It is the largest standard deviation that relative errors of size ε can produce, whatever their correlation. We adopt eq. (18), the more conservative form. It is the **model-error term**, and it is added in quadrature to the statistical uncertainties, as in eq. (13). The data cannot distinguish the two forms (table 2): both calibrate every degree of cancellation, with a correspondingly larger ε for eq. (17).

Why not simply inflate the noise? The obvious alternative, a systematic proportional to the statistical error, $\delta^2 = a(m)\, s^2_{\text{noise}}$, was used earlier in the project. It fixes the *average* calibration but makes the error the same fraction of the noise for every spectrum. It is therefore too small for the cancelling spectra and too large for the others (table 2, fig. 5c). The form eq. (18) uses the physics of eq. (16): it knows, from θ alone, which spectra cancel.

6.5 Measuring $\varepsilon(m)$ on data not used in the fit

The coefficient ε is determined on the 38,076 quick-test rows: single bins that were never used in the fit, about 480 per bin. In each bin k we solve

$$\left\langle \frac{(t-f)^2}{s_{\text{noise}}^2 + s_{\text{fit}}^2 + \varepsilon_k^2 S^2} \right\rangle_{\text{rows in bin } k} = 1, \qquad S = \sum_i |C_i(\theta) A_{ik}|, \tag{19}$$

for $\varepsilon_k \geq 0$ (by bisection; $\varepsilon_k = 0$ if the statistical errors alone already give $\langle z^2 \rangle \leq 1$, which happens in 5 of the 80 bins). Then $\ln \varepsilon$ is smoothed by a quadratic in $u = (m - 855\ \text{GeV})/600\ \text{GeV}$, fitted to the $\ln \varepsilon_k$ with weights proportional to the square root of the number of rows:

$$\ln \varepsilon(m) = p_0 + p_1 u + p_2 u^2. \tag{20}$$

These three numbers are the systematic's only parameters. They are stored with the model (`eps_poly` in `smooth_model.npz`) and evaluated by `SmoothHEFT.epsilon(m)`. Figure 5a shows the result: $\varepsilon = 0.077\%$ at 262 GeV, 0.28% at 600 GeV and 3.7% at 1.45 TeV. Near threshold, where the spectra are largest, the 23 components of POWHEG are consistent with one another to about one part in a thousand. That is a plausible size for the numerical precision of an NLO calculation.

6.6 Validation on the test spectra

The test spectra play no part in determining either the model or ε, so they give an independent check (table 2, fig. 5b,c).

	s.d. of the pull				$\langle z^2 \rangle$ by cancellation fifth
systematic	all	255–555	$>$ 555 GeV	$\lvert z \rvert < 2$	(1 = strongest … 5)
none	1.090	1.218	1.043	93.9%	1.70, 1.10, 1.07, 1.06, 1.01
$a(m)\, s_{\text{noise}}^2$	1.018	1.049	1.008	95.3%	1.45, 0.96, 0.94, 0.94, 0.90
$\varepsilon \sum_i \lvert C_i A_i \rvert$ **(adopted)**	**1.004**	**1.011**	**1.001**	**95.4%**	**0.97, 0.98, 1.04, 1.04, 1.00**
$\varepsilon\, [\sum_i (C_i A_i)^2]^{1/2}$	1.005	1.011	1.003	95.4%	1.01, 0.97, 1.03, 1.04, 1.00

Table 2: Calibration of the smooth model on the 1,000 test spectra, for four treatments of the systematic, each determined on the quick-test rows. For a standard normal, $|z| < 2$ holds for 95.4% of cases.

- With the model-error term the pulls have unit width overall, below and above 555 GeV, and in every fifth of cancellation. The most cancelling test spectrum goes from $\langle z^2 \rangle = 33.7$ to 1.9.
- On average the systematic is modest. It enlarges the total width by a median factor of only 1.014 over the noise (1.083 in the most cancelling fifth). The relative systematic on the cross section of a bin, $\delta/|f|$, has a median of 0.5% below 555 GeV and 1.5% overall, but reaches 20% or more in the 1% of bins with the strongest cancellation.
- The two lowest bins (255–285 GeV) remain slightly under-covered (pull s.d. 1.11 and 1.15, fig. 5b). The spectra rise steeply there, and finer knots at threshold may help.

6.7 Using the systematic

One bin. `model.spectrum(theta)` returns, for each bin, the prediction and $\sqrt{s_{\text{fit}}^2 + \delta^2}$; `discrepancy=False` gives s_{fit} alone. To compare with a POWHEG sample, add its Poisson noise.

Sums over bins. The model-error term is treated as independent between bins, but table 1 shows that the underlying errors are correlated over 5–8 bins. For a cross section summed over many bins the truth lies between two limits: independent bins, $\left(\sum_k \delta_k^2\right)^{1/2}$, and fully correlated bins, $\sum_k \delta_k$. For the SM in 255–1455 GeV:

$$\sigma_{\mathrm{SM}} = 28.11 \text{ fb}, \qquad s_{\mathrm{fit}} = 0.006 \text{ fb}, \qquad 0.031 \text{ fb} \leq \delta \leq 0.17 \text{ fb}. \tag{21}$$

The systematic exceeds the fit uncertainty by a factor of 5 to 30. It is still small (at most 0.6%), because the SM spectrum is only moderately cancelling overall ($R = 0.25$). Locally, however, the SM is strongly cancelling near threshold, where the box and triangle diagrams interfere destructively: the relative systematic is 5.5% in the 255–270 GeV bin, against 0.6% at 600 GeV.

In a likelihood. In an analysis the systematic can be represented as a nuisance parameter per bin (or per group of correlated bins), with a Gaussian constraint of width δ_k, in the usual way. This is the frequentist counterpart of the "model discrepancy" term of Kennedy and O'Hagan [8], which adds to a simulator's output a random function describing the simulator's own inadequacy.

6.8 Where does it come from? Open questions

- **The generator.** By the argument of section 6.3, the departures are smooth in m_{hh} within a run and vary from run to run. A likely candidate is the numerical evaluation of the two-loop virtual amplitudes, which in the POWHEG `ggHH` implementation are interpolated from pre-computed grids, together with each run's own Monte Carlo integration. Questions for the colleagues who produced the samples: were the HEFT points independent runs with independent seeds and grids? What relative precision is quoted for the virtual amplitudes? Were there negative-weight events? Regenerating a few strongly cancelling spectra at higher precision would settle the matter.
- **Or the form.** Alternatively, some ingredient of the generation could depend on θ (eq. (6) and the box that follows it), in which case the 23-term form itself would be incomplete for these samples. A diagnostic: such a departure would vary smoothly with θ, so the residual shapes of test spectra close together in θ would be similar. Run-to-run numerical errors would make them unrelated.
- **The rise at high mass.** ε grows to 3.7% at 1.45 TeV, where the bins hold about 20 events. Whether this reflects the generator's precision at high mass, the overflow above 1500 GeV, or the limits of measuring ε from so few events is open. It matters little for the cross section, which is small there.
- **Its effect on the a_i.** Refitting with δ^2 added to the variances ("second pass") leaves the calibration unchanged (pull s.d. 1.003) but fits the cancelling spectra less closely ($-2\ln\lambda$ near threshold 1.87 against 1.57). Since the purpose of the model is to reproduce POWHEG, the first pass is the primary model. The difference between the passes measures how much the individual a_i depend on the treatment of the cancelling spectra.
- **Improvements.** A Gaussian-process systematic with amplitude δ and a correlation length fitted to the residuals would replace the two limits above with one number for any sum over bins.

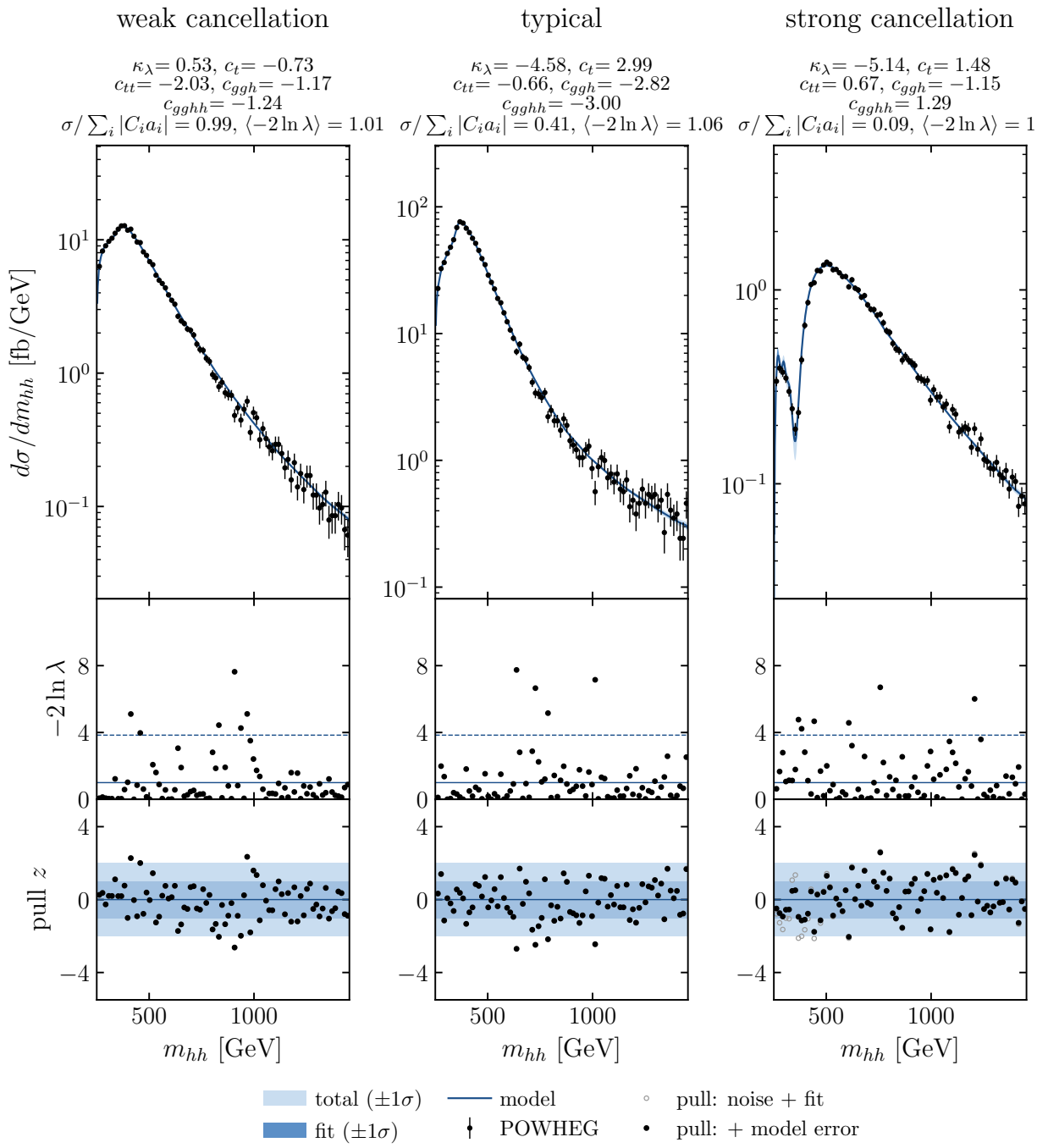


Figure 4: Three of the 1,000 test spectra, chosen by how strongly their 23 terms cancel: weak ($\sigma / \sum_i |C_i a_i| \approx 1$), typical and strong; in each case the spectrum with the median mean $-2 \ln \lambda$ within its fifth of the cancellation distribution. Top: POWHEG (points) and the smooth model with its fit and total ($\pm 1\sigma$) bands. Middle: the likelihood-ratio statistic of each bin, $-2 \ln \lambda = -2 \ln[\text{Poisson}(n; \mu) / \text{Poisson}(n; n)]$, with n the POWHEG event count and μ the model prediction; for a correct model it follows χ^2_1, with mean 1 (solid line) and 95% point 3.84 (dashed). Bottom: the pull $z = (n - \mu)/\text{sd}$ with sd the total uncertainty (filled) and without the model-error term (open); the bands mark $|z| < 1$ and $|z| < 2$. The likelihood-ratio statistic measures the quality of the prediction; the pull, the quality of its uncertainty.

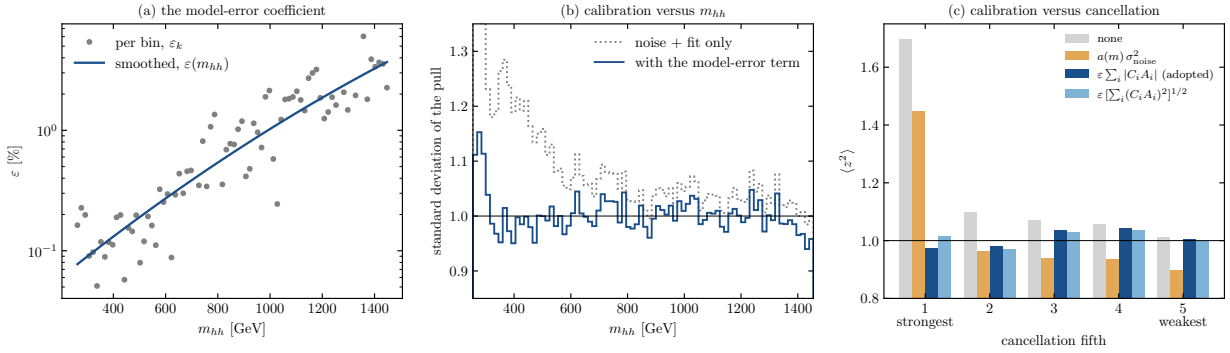


Figure 5: The systematic uncertainty. (a) The model-error coefficient: per bin, from the quick-test rows (points), and its three-parameter smooth form (line). (b) Standard deviation of the pull eq. (13) on the 1,000 test spectra in each bin, without (dotted) and with (solid) the model-error term. (c) $\langle z^2 \rangle$ by fifth of the cancellation ratio eq. (15), for no systematic, a systematic proportional to the noise, and the two forms eq. (18) (adopted) and eq. (17). Each systematic was determined on the quick-test rows; 1 is perfect calibration.

7 Accuracy and other checks

model	mean $-2\ln\lambda$ per bin 255–555 GeV	> 555 GeV	median per spectrum	parameters
smooth model	**1.566**	**1.100**	**1.078**	946
free fit per bin (benchmark)	1.559	1.105	1.075	1,840
network ensemble (cross-check)	1.561	1.121	1.085	$20 \times 3{,}944$

Table 3: Accuracy on the 1,000 test spectra; 1 is the value for a perfect model with Poisson noise only. The common excess below 555 GeV is the systematic of section 6.

- **Accuracy** (table 3). The smooth model is as accurate as the free fit with half as many parameters, and slightly better than the networks above 555 GeV, where the networks were trained with observed-count weights and are biased low.
- **Inside a bin: the merged-bin test.** Fitted to the training spectra merged into 30 GeV bins, the model predicts how each 30 GeV bin of the test spectra splits into two 15 GeV bins to within the noise above about 360 GeV ($\chi^2 \approx 1$). The free fit can only halve each value, and is up to 30 times worse. Below 345 GeV the threshold rise has structure finer than 30 GeV.
- **Between bin centres.** The model is a density, defined at every m. The networks, by contrast, are constrained only at the bin centres, and below 400 GeV their spread between the centres is 20–90 times larger than at them (`docs/paper/fig_interpolation.pdf`).
- **Sensitivity.** $J = 30$, 40 and 60 give the same accuracy and calibration to within 0.01; the evidence prefers $J = 40$.
- **Range.** Predicting the unused bins, 240–255 GeV comes out 8% high, while 1455–1500 GeV is unreliable (the splines there are fixed by the prior alone). **Use the model on 250–1455 GeV only.**
- **The coefficient functions** (fig. 6) are smooth, with exact bands, and agree with the networks except for the least constrained ones. a_{16}–a_{20}, whose monomials are small or nearly degenerate over the sampled θ, are determined to 20–75%. Their uncertainty is large but exact, and it propagates correctly into every prediction.

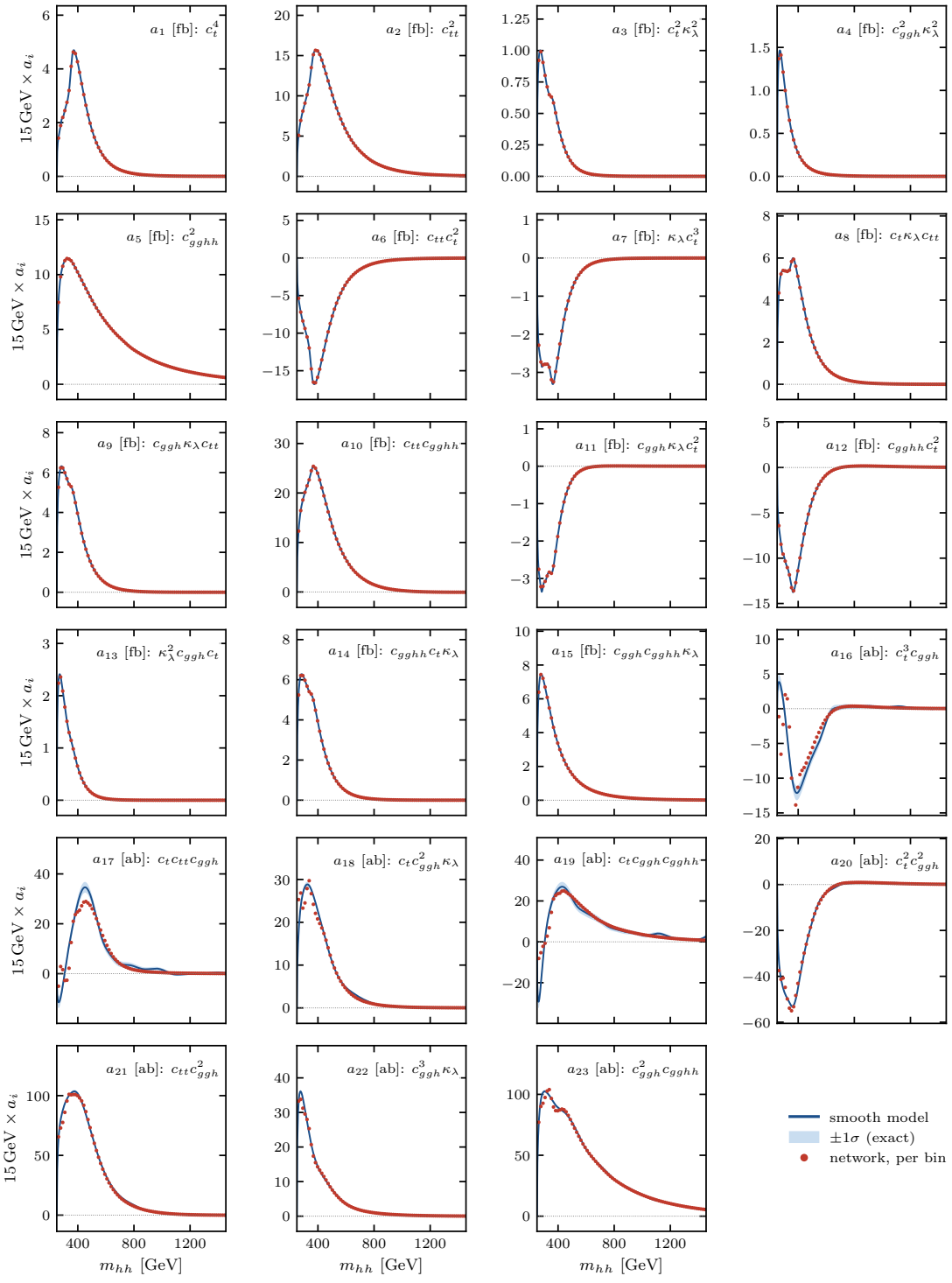


Figure 6: The 23 coefficient functions $a_i(m_{hh})$ of the smooth model, shown as $15\,\mathrm{GeV} \times a_i$, the cross section per 15 GeV multiplying the monomial $C_i(\theta)$ in each panel, so that $d\sigma/dm_{hh} = \sum_i C_i(\theta)\, a_i(m_{hh})$. Line: posterior mean; band: exact $\pm1\sigma$; red points: the per-bin values of the network ensemble, for comparison. Units are fb, or ab for the eight smallest functions. The functions whose monomials are small or nearly degenerate over the sampled HEFT points (a_{16}–a_{20}) are the least well determined.

8 Using the model

The fitted model is a single file, `code/runs/SMOOTH_J40/smooth_model.npz` (about 7 MB: the breakpoints, $\hat{\beta}$, H, the λ_i and the three numbers of ε). It is written by `code/heft_smooth.ipynb` ($\approx$ 75 s on a laptop CPU). Using it needs only NumPy and SciPy, not PyTorch:

```
import numpy as np
from dihiggs.smooth import SmoothHEFT, heft_monomials

model = SmoothHEFT.load('runs/SMOOTH_J40/smooth_model.npz')

theta = [[1.0, 1.0, 0.0, 0.0, 0.0],      # (klambda, ct, ctt, cggh, cgghh): the SM
         [2.0, 1.0, 0.0, 0.0, 0.0]]      # one row per HEFT point

# cross section (pb) in the 80 bins of 15 GeV from 255 to 1455 GeV, and its
# uncertainty: fit uncertainty and model-error term, added in quadrature
sigma, sd = model.spectrum(theta)                    # arrays [2, 80]
sigma, sd_fit = model.spectrum(theta, discrepancy=False)

# any other binning within 250-1455 GeV
sigma10, sd10 = model.spectrum(theta, edges=np.arange(260, 1451, 10))

# the density d sigma/dm (pb/GeV) at any masses, with its fit uncertainty
m = np.linspace(255, 1455, 1201)
dens, dens_sd = model.dsigma_dm(theta, m)

# the 23 coefficient functions a_i(m) with their bands, and epsilon(m)
a, a_sd = model.coefficients(m)                      # arrays [len(m), 23]
eps = model.epsilon(m)                               # a fraction: 0.00077 at threshold
C = heft_monomials(theta)                            # the 23 monomials C_i(theta)
```

Wide bins and the model-error term

`spectrum` evaluates ε at the centre of each bin. That is right for bins of 15 GeV or narrower. For a wide bin it is not: one bin from 255 to 1455 GeV gives the SM 28.11 ± 0.73 fb, because $\varepsilon(855\ \text{GeV}) = 0.65\%$ is applied to terms that come mostly from near threshold, where ε is up to eight times smaller. For wide intervals, sum 15 GeV bins and combine the δ_k with the limits of section 6.7 (here 0.03–0.17 fb).

9 Cross-checks: the network and the Bayesian network

The first version of this project modelled the 23 functions with a neural network, `HEFTNet` (`dihiggs/heftnet.py`, fig. 7). The network keeps the exact polynomial in θ and learns only $a_i(m_{hh}) = P_i(m_{hh})\, e^{Q_i(m_{hh})}$. P is a six-layer ReLU network with a tanh output, which supplies shape and sign. Q is one linear layer, which supplies the overall size and an exponential fall-off. The network is trained with an L1 loss on the bin values at the bin centres, then fine-tuned with a Huber loss of the pulls. An ensemble of 20 networks (bootstrap of the training spectra) reaches the same accuracy floor as the smooth model (table 3). With the same model-error term its pulls are calibrated equally well (s.d. 1.004).

A Bayesian treatment of the network was also studied. Hamiltonian Monte Carlo could not

explore the posterior, whose curvatures span many orders of magnitude, and gave bands of only 0.5% of the noise. A mixture of linearised Laplace approximations about the 20 networks gives 2.4% of the noise within and 5.6% between networks (`docs/bnn/laplace_explained.pdf`). Like the smooth model, it needs the model-error term to be calibrated. This is the main lesson of the cross-checks: Bayes' theorem works, the parameter uncertainty of any 23-term model is genuinely small, and the dominant uncertainty is the systematic of section 6, which no parameter posterior of a 23-term model can contain.

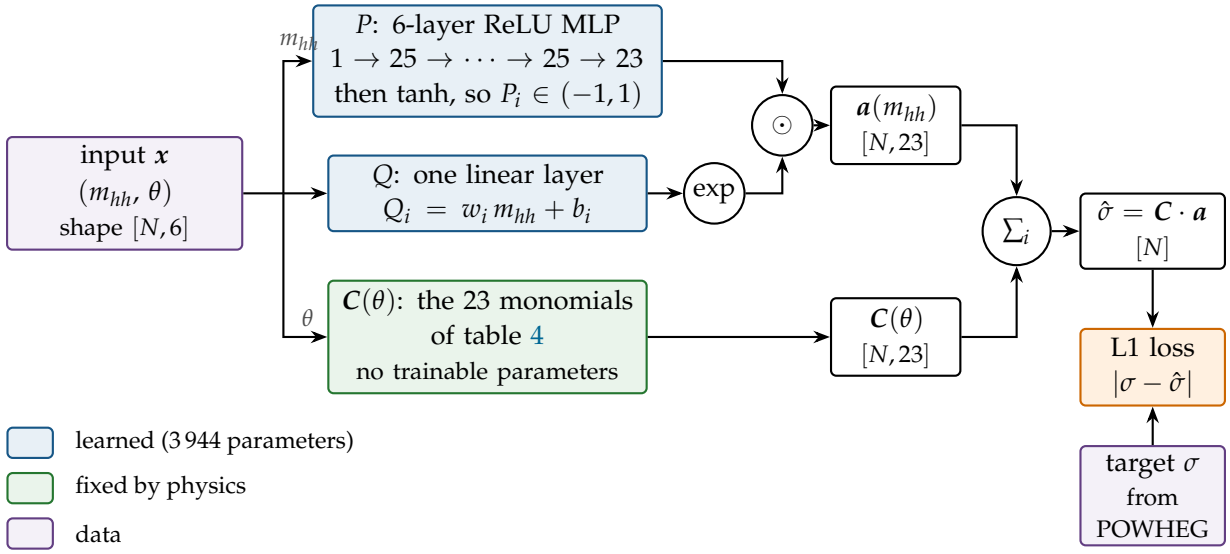


Figure 7: The `HEFTNet` model. Each input row, $x = (m_{hh}, \theta)$ with $\theta = (\kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh})$, is split. The mass m_{hh} feeds two small networks, P and Q, which give the 23 functions $a_i(m_{hh}) = P_i(m_{hh})\, e^{Q_i(m_{hh})}$. The Wilson coefficients θ feed the 23 known monomials $C_i(\theta)$. The prediction is the dot product $C \cdot a$, which is compared with the POWHEG cross section using the absolute (L1) loss. Blue boxes are learned; the green box is fixed by physics.

10 Points worth noticing

- **Positivity is not enforced.** Nothing forces $d\sigma/dm \geq 0$. On the test spectra all 80,000 bin predictions are positive, but near deep interference dips at other θ it is worth checking.
- **Identifiability.** At fixed m, the density is linear in the 23 unknowns a_i. They are determined only if the 23 monomials, evaluated at the training points, are linearly independent. With thousands of widely spread θ points they are, although some combinations (a_{16}–a_{20}) only weakly. With a few points close to the SM they would not be.
- **Correlations.** The posterior covariance H^{-1} correlates every prediction with every other. For a single bin this does not matter. For sums over bins, or for joint fits of several spectra, use H^{-1} through `SmoothHEFT.predict` with the summed basis integrals, rather than adding per-bin uncertainties.
- **Reproducibility.** There is no training: the same data give the same model to machine precision, and the paper figures are regenerated from `runs/SMOOTH_J40/smooth_model.npz` by the scripts in `docs/paper/` (shared code in `smooth_common.py`).

A The 23 monomials

i	$C_i(\theta)$	origin	i	$C_i(\theta)$	origin
1	c_t^4	$v_1 v_1$	13	$\kappa_\lambda^2 c_{ggh} c_t$	$v_2 v_4$
2	c_{tt}^2	$v_3 v_3$	14	$c_{gghh} c_t \kappa_\lambda$	$v_2 v_5$
3	$c_t^2 \kappa_\lambda^2$	$v_2 v_2$	15	$c_{ggh} c_{gghh} \kappa_\lambda$	$v_4 v_5$
4	$c_{ggh}^2 \kappa_\lambda^2$	$v_4 v_4$	16	$c_t^3 c_{ggh}$	$(c_t c_{ggh})\, v_1$
5	c_{gghh}^2	$v_5 v_5$	17	$c_t c_{tt} c_{ggh}$	$(c_t c_{ggh})\, v_3$
6	$c_{tt} c_t^2$	$v_1 v_3$	18	$c_t c_{ggh}^2 \kappa_\lambda$	$(c_t c_{ggh})\, v_4$
7	$\kappa_\lambda c_t^3$	$v_1 v_2$	19	$c_t c_{ggh} c_{gghh}$	$(c_t c_{ggh})\, v_5$
8	$c_t \kappa_\lambda c_{tt}$	$v_2 v_3$	20	$c_t^2 c_{ggh}^2$	$c_{ggh}^2\, v_1$
9	$c_{ggh} \kappa_\lambda c_{tt}$	$v_3 v_4$	21	$c_{tt} c_{ggh}^2$	$c_{ggh}^2\, v_3$
10	$c_{tt} c_{gghh}$	$v_3 v_5$	22	$c_{ggh}^3 \kappa_\lambda$	$c_{ggh}^2\, v_4$
11	$c_{ggh} \kappa_\lambda c_t^2$	$v_1 v_4$	23	$c_{ggh}^2 c_{gghh}$	$c_{ggh}^2\, v_5$
12	$c_{gghh} c_t^2$	$v_1 v_5$			

Table 4: The monomials in the order used in `heftnet.py` and `smooth.heft_monomials`, which is also the order of Eq. (4.1) of Ref. [1]. Terms 1–15 are the products $v_i v_j$ of the LO coupling factors in eq. (3) and suffice at LO. Terms 16–23 appear at NLO. Their "origin" column shows them as an NLO coupling factor ($c_t c_{ggh}$ or c_{ggh}^2) times an LO factor v_i, as described in section 2. In the code, `klambda` $= \kappa_\lambda$, `CT` $= c_t$, `CTT` $= c_{tt}$, `CGGH` $= c_{ggh}$, `CGGHH` $= c_{gghh}$.

References

[1] L. Alasfar *et al.*, *Effective Field Theory descriptions of Higgs boson pair production*, LHC Higgs Working Group public note, arXiv:2304.01968 (in this repository as `docs/theory.pdf`).

[2] N. Dalmasso, L. Masserano, D. Zhao, R. Izbicki and A. B. Lee, *Likelihood-free frequentist inference: bridging classical statistics and machine learning for reliable simulator-based inference*, arXiv:2107.03920.

[3] A. Al Kadhim *et al.*, *Amortized simulation-based frequentist inference for tractable and intractable likelihoods*, Mach. Learn.: Sci. Technol. (2024), doi:10.1088/2632-2153/ad218e.

[4] C. de Boor, *A Practical Guide to Splines*, Springer (1978).

[5] P. H. C. Eilers and B. D. Marx, *Flexible smoothing with B-splines and penalties*, Statist. Sci. **11** (1996) 89.

[6] D. J. C. MacKay, *Bayesian interpolation*, Neural Comput. **4** (1992) 415.

[7] S. Baker and R. D. Cousins, *Clarification of the use of chi-square and likelihood functions in fits to histograms*, Nucl. Instrum. Meth. **221** (1984) 437.

[8] M. C. Kennedy and A. O'Hagan, *Bayesian calibration of computer models*, J. R. Statist. Soc. B **63** (2001) 425.